**A Formal Physics Laboratory Report**

**REPORT LAB 07**

# DC AND AC CIRCUITS

| **Class: CC09** **Group: 10** | **Lecturer' s comment** |
|---|---|
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## I. Aims/Purposes:

- Solidifying the concept DC and AC circuits; verifying the Stefan's law and the Ohm's law with an incandescent light bulb; determining the unknown resistance, capacitance, and inductances.
- Be acquainted with the digital multimeter to determine the voltage as well as the amperage in the DC and AC circuit or the resistance of conductor.
- Investigating the dependence of the resistance of the metal by plotting the volt-ampere characteristic of the bulb's filament,therefore, determine the temperature of the bulb's filament.

## II. Method, Equipment and Procedure:

*1) <u>Equipment</u>*

- 2 digital multimeters
- Incandescent light bulb 12V-3W
- Unknown-value Resistor
- Unknown-value Capacitor
- Unknown-value Inductors
- Variable power supply 12V-3A/AC-DC
- Plug in board

*2) <u>Method</u>*

- Indirect measurement

*3) <u>Procedure</u>*

*3.1) Determining the filament temperature of an incandescent light bulb and checking the operation of the variable power supply 12V-3A*

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- First step, we need to connect the digital multimeters to the light bulb. Remember to adjust the needle to 200 ohm so that we can find the temperature of the light bulb in room condition.
- Then we connect the power supply to the plug in board in Direct Current. We use the digital multimeter to find the voltage of the light bulb, another digital multimeter we use to measure the ampere value of the light bulb. After connection, turn on the supply power and rotate the dial as you witness the light bulb is on. So you can find the value of the voltage of the light bulb

*3.2) Graphing characteristic volt-ampere of the tungsten filament*

- Secondly, we adjust the value from 1 to 10 by rotating the small dial. For each voltage value, we have a different ampere data.
- After collecting 10 values of I, we have to sketch the graph

*3.3) Determining the capacitance of the capacitor in the RC circuit and the inductance L of the windings in the RL circuit*

- Third step, we connect the power supply to the plug in board in Alternating Current. Remember to connect the ampere meter.
- After that, we have 4 cases. Case 1: R series L; case 2: R series $C_1$; case 3: R series $C_1$ series $C_2$; case 4: R series ($C_1$ parallels $C_2$)
  - First case: R series L, do not forget to take the ampere value. Then we need to find U(R), U(L) and U(RL)
  - Second case: R series $C_1$, we need to replace the induction coil with capacitance. Then, take the new value of the ampere meter. Find U, U(R), U($C_1$), U(R$C_1$)
  - Third case: R series $C_1$ series $C_2$, we need to replace the position of wires and remember to take the new ampere's value. Then we find the value of U, U(R), U($C_1C_2$) and U(R$C_1C_2$)
  - Last case: R series ($C_1$ parallels $C_2$), we also need to change the position of wires and remember to take the new ampere's value. Then we find the value of U, U(R), U($C_1$ // $C_2$) and U(R$C_1C_2$)

## III. Equations

*1) Calculating $R_0$, T*

$$R_0 = \frac{R_p}{1+\alpha t_p+\beta t_p^{\ 2}}$$

$$T = 273 + \frac{1}{2\beta}\left[\sqrt{\alpha^2 + 4\beta\left(\frac{R_t}{R_0} - 1\right)} - \alpha\right]$$

*2) Calculation of the capacitance values of these capacitors, and the inductance of the coil.*

$$C = \frac{1}{2\pi f Z_C} = \frac{1}{2\pi f U_C}$$

$$L = \frac{\sqrt{Z_{cd}^{\ 2} - r^2}}{2\pi f}$$

*3) Calculation of the uncertainties*

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$$\Delta T = \frac{1}{2}\left[\left(\frac{4\beta}{\alpha^2 R_0 + 4\beta(R_t - R_0)}\Delta R_t + \left(\frac{\alpha^2 - 4\beta}{\alpha^2 R_0 + 4\beta(R_t - R_0)}\right)\Delta R_0\right)\right] \times \sqrt{\alpha^2 + 4\beta\left(\frac{R_t}{R_0} - 1\right)}$$

$$\frac{\Delta R_0}{R_0} = \frac{\Delta R_p}{R_p} + \left(\frac{\alpha + 2\beta t_p}{1 + \alpha t_p + \beta t_p^{\,2}}\right)\Delta t_p$$

$$\frac{\Delta C}{C} = \frac{\Delta U_C}{U_c} + \frac{\Delta I}{I} + \frac{\Delta \pi}{\pi} + \frac{\Delta f}{f}$$

$$\frac{\Delta L}{L} = \frac{Z_{cd}\Delta Z_{cd} + r\Delta r}{Z_{cd}^{\,2} - r^2} + \frac{\Delta \pi}{\pi} + \frac{\Delta f}{f} \; and \; \frac{\Delta Z_{cd}}{Z_{cd}} = \frac{\Delta U_L}{U_L} + \frac{\Delta I}{I}$$

## IV. Experimental data

**Table 1**

| Voltmeter DC<br>$U_m$= 20V α = 0.01<br>δU= 0.5% n = 3 | | | Ammeter DC<br>$I_m$= 200mA α= 0.1<br>δI= 1.2% n= 5<br>$t_p$= 30 ± 1°C | | | Ohmmeter<br>$R_m$= 200Ω α= 0.1<br>δR= 1.0% n= 3<br>$R_p$= 7Ω | |
|---|---|---|---|---|---|---|---|
| U (V) | ΔU (V) | I (mA) | ΔI (mA) | U (V) | ΔU (V) | I (mA) | ΔI (mA) |
| 1 | 0.035 | 56.3 | 1.10 | 6 | 0.060 | 133.2 | 2.10 |
| 2 | 0.040 | 73.6 | 1.24 | 7 | 0.065 | 146.0 | 2.25 |
| 3 | 0.045 | 92.3 | 1.41 | 8 | 0.070 | 158.6 | 2.40 |
| 4 | 0.050 | 108.5 | 1.56 | 9 | 0.075 | 169.4 | 2.53 |
| 5 | 0.055 | 122.4 | 1.71 | 10 | 0.080 | 180.1 | 2.66 |

**Table 2**

| | Voltmeter AC<br>$U_m$= 20V α= 0.01<br>δU= 1.0% n= 5 | | | | Ammeter AC:<br>$I_m$= 200mA α= 0.1mA<br>δI= 1.8% n= 3 | | | |
|---|---|---|---|---|---|---|---|---|
| | I (mA) | U (V) | $U_R$ (V) | $U_C$ (V) | Z | R | $R_C$ | C |
| $C_1$ | 34.5 | 12.40 | 10.09 | 5.05 | 359.42 | 292.46 | 145.51 | $2.19 \times 10^{-5}$ |
| Serial capacitor circuit $C_1$ , $C_2$ | 27.4 | 12.35 | 8.82 | 8.11 | 450.73 | 321.9 | 295.9854 | $1.08 \times 10^{-5}$ |
| Parallel capacitor circuits $C_1$,$C_2$ | 37.1 | 12.32 | 11.63 | 2.73 | 332.07 | 313.473 | 73.585 | $4.33 \times 10^{-5}$ |

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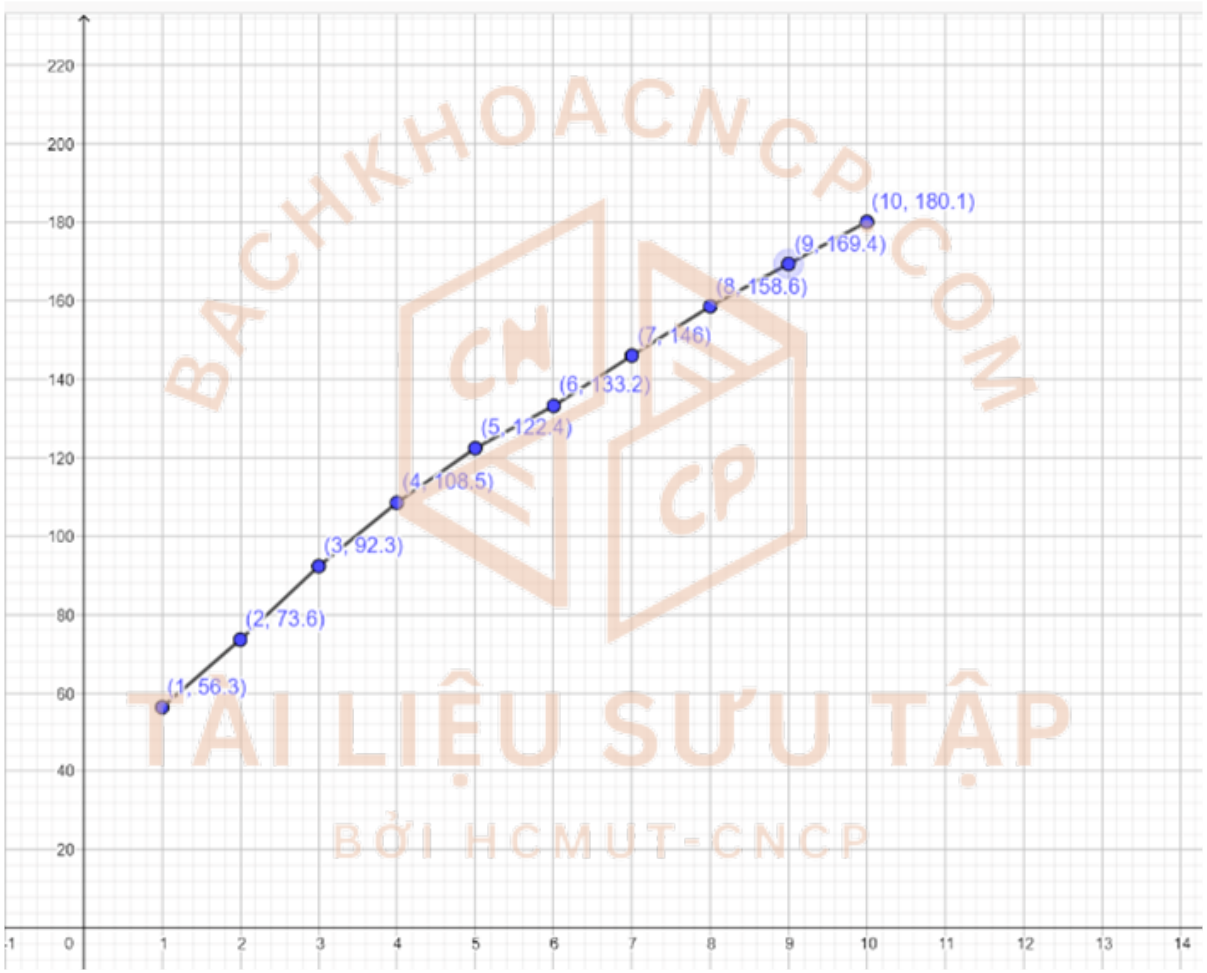
**Table 3**

| Ohmmeter: $R_m$= 200Ω | | α= 0.1 | | δr = 1.0% | | n = 3 | |
|---|---|---|---|---|---|---|---|
| I (mA) | U (V) | $U_R$(V) | $U_L$(V) | Z | R | $Z_L$ | L |
| 25.4 | 12.30 | 8.03 | 6.26 | 484.252 | 316.1417 | 246.46 | 0.75 |

## V. Calculation works

### V.1. Plot the volt-ampere characteristic of the bulb's filament




### V.2. Calculating $R_0$, T when U = 10V

$$R_0 = \frac{R_p}{(1+\alpha t_p+\beta t_p^2)} = \frac{7\ 10^{-33}}{(1+4.82.10^{-3}\ .(30+273)\ +6.76.10^{-7}(30+273)2} = 2.775(\Omega)$$

$$R = \frac{U}{I} = \frac{10}{200.10^{-3}} = 50(\Omega)$$

$$T = 273 + \frac{1}{2\beta}\left[\sqrt{\alpha^2 + 4\beta\left(\frac{R_t}{R_0} - 1\right)} - \alpha\right]$$

$$273 + \frac{1}{2.676.10^{-7}}\left[\sqrt{(4.82.10^{-3})^2 + 4.676.10^{-77}\left(\frac{50}{2.775} - 1\right)} - 4.82.10^{-3}\right] = 2862.945(K)$$

**V.3. Calculation of the capacitance values of these capacitors, and the inductance of the coil**

$$C = \frac{1}{2\pi f Z_c} = \frac{1}{2\pi f U_c} = \frac{34.5.10^{-3}}{2\pi.50.5.02} = 2.18910^{-5}(F)$$

***2. Serial capacitor circuit $C_1$ , $C_2$***

$$C = \frac{1}{2\pi f Z_c} = \frac{1}{2\pi f U_c} = \frac{27.4.10^{-3}}{2\pi.50.8.11} = 1.076.10^{-5}(F)$$

*3.* ***Parallel capacitor circuits $C_1$, $C_2$***

$$L = \frac{\sqrt{Z_{cd}^2 - r^2}}{2\pi f} = \frac{\sqrt{246.457^2 - 72^2}}{2\pi.50} = 0.751(H)$$

$$Z_{cd} = \frac{U_L}{I} = \frac{6.26}{25.410^{-3}} = 246.457\ (V)$$

$$C = \frac{1}{2\pi f Z_c} = \frac{1}{2\pi f U_c} = \frac{37.1.10^{-3}}{2\pi.50.2.73} = 4.328.10^{-5}(F)$$

**V.4. Calculation of the uncertainties**

1. **$C_1$**

$$\delta = \frac{\Delta C1}{U_{C1}} = \frac{\Delta C1}{U_{C1}} + \frac{\Delta I}{I} + \frac{\Delta\pi}{\pi} + \frac{\Delta f}{f} = \frac{5{,}02.1\%.+5..0.01}{5.02} + \frac{34.5.0.8\%+3.01}{34.5} + \frac{0.005}{3.14} + \frac{0.5}{50}$$

$=0.058248$

$$\Delta C1 = C1.\delta = 0.058248.2.18910^{-5} = 1.275.10^{-6}(F)$$

2. ***Serial capacitor circuits $C_1$,$C_2$***

$$\delta = \frac{\Delta C1}{U_{C1}} = \frac{\Delta C1}{U_{C1}} + \frac{\Delta I}{I} + \frac{\Delta\pi}{\pi} + \frac{\Delta f}{f} = \frac{8.11.1\%.+5..0.01}{8.111} + \frac{27.4.1..\%+3.01}{27.4} + \frac{0.005}{3.14} + \frac{0.5}{50}$$

$=0.056706$

$$\Delta C1 = C1.\delta = 0.056706.1.076.10^{-5} = 6.102.10^{-7}(F)$$

3. ***Parallel capacitor circuits $C_1$, $C_2$***

$$\delta = \frac{\Delta C2}{U_{C2}} = \frac{\Delta C2}{U_{C2}} + \frac{\Delta I}{I} + \frac{\Delta\pi}{\pi} + \frac{\Delta f}{f} = \frac{2.73.1\%.+5..0.01}{2.73} + \frac{37.1.1.8\%+3.01}{37.1} + \frac{0.005}{3.14} + \frac{0.5}{50}$$

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$=0.065994$

$\Delta C2 = C2.\delta = 0.065994.4.328.10^{-5} = 2.856.10^{-6}(T)$

$\delta = \frac{\Delta Z_{cd}}{Z_{cd}} = \frac{\Delta U_L}{U_L} + \frac{\Delta I}{I} = \frac{6.26.1\%.+5..0.01}{6.21} + \frac{25.4.1.8.\%+3.0.1}{25.4} = 0.048$

$\Delta Z_{cd} = C1 . Z_{cd} = 0.048.246.457 = = 11.830(\mathrm{H})$

$\Delta r$ **= 72 .**$1\% + 3.0.1 = 1.02\ \Omega$

$\delta = \frac{\Delta L}{L} = \frac{Z_{cd}\Delta Z_{cd} + r\Delta r}{Z_{cd}^{2} - r^{2}} + \frac{\Delta\pi}{\pi} + \frac{\Delta f}{f} = \frac{246.457 \bullet 11.830 + 72 \bullet 1.02}{246.457^{2} - 72^{2}} + \frac{0.005}{3.14} + \frac{0.5}{50} = 0.065$

$\Delta L = L.\delta = 0.065.0.751 = 0.049\ (H)$

$\Delta R_p = 7 . 1\% + 3 . 0.1 = 0.37\ \Omega$

$\frac{\Delta R_0}{R_0} = \frac{\Delta R_p}{R_p} + \left(\frac{\alpha + 2\beta t_p}{1 + \alpha t_p + \beta t_p^{2}}\right)\Delta t_p =$

$\frac{0.37}{7} + \left(\frac{4.812.10^{-3}.6.76.10^{-7}.(30+273).2}{1 + 4.82.10^{-3}.(30+273) + 6.76.10^{-7}..(30+273)^{2}}\right)(1 + 273) = 0.621$

$\Delta R_0 = 0.621.2.775 = 1.723\ \Omega$

$\frac{\Delta R_t}{R_t} = \frac{\Delta U}{U} + \frac{\Delta I}{I} = \frac{0.080}{10} + \frac{2.661}{180.1} = 0.0228$

$\Delta R_t = 0.0228 . 50 = 1.14\ \Omega$

$\Delta T = \frac{1}{2}\left[\left(\frac{4\beta}{\alpha^{2}R_0 + 4\beta(R_t - R_0)}\right)\Delta R_t + \left(\frac{\alpha^{2} - 4\beta}{\alpha^{2}R_0 + 4\beta(R_t - R_0)}\right)\Delta R_0\right].\sqrt{\alpha^{2} + 4\beta\left(\frac{R_t}{R_0} - 1\right)}$

$= 8.326.10^{-4}$

**V.5. Results**

**Resistor**

| |
|---|
| $R = \overline{R_0} \pm \Delta R_0 = 2.775 \pm 1.723(\Omega)$ |
| $T = \overline{T_0} \pm \Delta T_0 = 2862.945 \pm 8.326 . 10^{-4}(°\mathrm{C})$ |

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**Capacitors**

$C = \overline{C} \pm \Delta C = 21.89.10^{-6} \pm 1.275.10^{-6}$ (F)

$C_1 = \overline{C1} \pm \Delta C_1 = 10.76.10^{-6} \pm 0.610.10^{-6}$ (F)

$C_2 = \overline{C2} \pm \Delta C_2 = 43.28.10^{-6} \pm 2.856.10^{-6}$ (F)

**Inductor**

$r = \overline{r} \pm \Delta r = 72 \pm 1.02$ (Ω)

$L = \overline{l} \pm \Delta L = 0.751 \pm 0.049$ (H)

**VI/ Questions**

**Question 1: State and write the equation of Ohm's law for a constant current. Why is the characteristic volt-ampere (I = f(U)) of an incandescent light bulb not a straight line?**

According to Ohm's law for a DC circuit, the current I flowing through the circuit is proportional to the potential difference U between the two ends of the circuit and inversely proportional to the resistance R of the circuit:

$$I = \frac{U}{R} \qquad (5)$$

The Characteristic volt-ampere (I = f(U)) of an incandescent light bulb is not a straight line because the conductor's resistance varies with the voltage applied to the two ends of the conductor and does not obey Ohm's law. U and I differ in a non-proportional way. A light bulb's filament resistance varies with temperature.

**Question 2: State the relationship between frequency, phase, and amplitude of alternating current and alternating voltage in a circuit:**
**- Resistance R;**
**- Capacitance C;**
**- Inductance L.**

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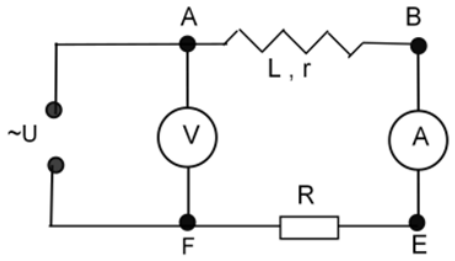


*Figure Lab 1-5: RL circuit schematic*

Suppose an alternating current flowing in the circuit at time $t$ has the form:

$$i = I_o.\sin 2\pi f\,.t \qquad (21)$$

at that time:

$$u = u_R + u_r + u_L \qquad (22)$$

$u_R$ and $u_r$ are in same phase with $i$, $u_L$ is early than $i$ by a phase of π/2:

$$u = U_{0R}.\sin 2\pi f.t + U_{0r}.\sin 2\pi f.t + U_{0L}.\sin(2\pi f.t - \pi\frac{\pi}{2}) \quad (13)$$

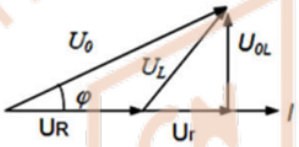


*Figure Lab 1-6: The phasor diagram of RL circuit*

Using the phasor diagram (or Fresnel vector diagram) of RL circuit as shown in Figure Lab 1-6, we will find the alternating voltage u:

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$$u = U_{0R}.\sin(2\pi f.t + \varphi) \qquad (23)$$

with

$$U_0 = \sqrt{(U_{0R} + U_{0r})^2 + U_{0L}^2} \qquad (24)$$

$$tg\varphi = \frac{U_{0L}}{U_{0R}+U_{0r}} = \frac{2\pi fL}{R+r} \qquad (25)$$

Substitute $U_{0R} = I_0R$, $U_{0r} = I_0r$ và $U_{0L} = I_0Z_L = I_0 2\pi fL$ into Eq. 24:

$$U_0 = I_0\sqrt{(R+r)^2 + Z_L^2} = I_0Z \qquad (26)$$

with $Z_L$ is the inductance of the inductor:

$$Z_L = 2\pi fL \qquad (27)$$

The total impedance $Z$ of the RL circuit:

$$Z = \sqrt{(R+r)^2 + Z_L^2} \qquad (28)$$

Dividing both sides of Eq. 26 by $\sqrt{2}$, we have Ohm's law for the RC circuit as:

$$I = \frac{U}{Z} \qquad (29)$$

with $U$ and $I$ are the r.m.s. values of the AC voltage and current in the RC circuit, that can be measured with a digital multimeter.

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**Question 3: Using a Fresnel vector diagram, establish a frequency, phase, and amplitude relationship between AC amperage and AC potential difference in an unbranched RLC circuit. Since then infer the expression determined impedance of the RLC circuit. Conditions for amperage reach the peak in RLC circuit?**

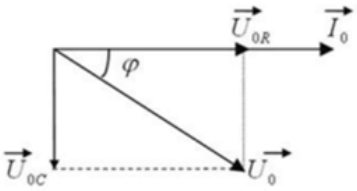


***Figure Lab 1-4: The phasor diagram of RC circuit***

Using the phasor diagram (*or Fresnel vector diagram*) of RC circuit as shown in Figure Lab 1-4, we will find the alternating voltage *u*:

$$u = U_{0R}.\sin(2\pi f.t + \varphi) \qquad (14)$$

with

$$U_0 = \sqrt{U_{0R}^2 + U_{0C}^2} \qquad (15)$$

$$tg\varphi = -\frac{U_{0C}}{U_{0R}} = -\frac{1}{2\pi fCR} \qquad (16)$$

Substitute $U_{0R} = I_0R$ và $U_{0C} = I_0Z_C = \frac{I_0}{2\pi fC}$ into Eq. 15:

$$U_0 = I_0\sqrt{R^2 + Z_C^2} = I_0Z \qquad (17)$$

with $Z_C$ is the capacitance of the capacitor:

$$Z_C = \frac{1}{2\pi fC} \qquad (18)$$

The total impedance *Z* of the RC circuit:

$$Z = \sqrt{R^2 + Z_C^2} \qquad (19)$$

Dividing both sides of Eq. 17 by $\sqrt{2}$, we have Ohm's law for the RC circuit as:

$$I = \frac{U}{Z} \qquad (20)$$

with *U* and *I* are the root-mean-square (*r.m.s.*) values of the AC voltage and current in the RC circuit, that can be measured with a digital multimeter.

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**Question 4: Demonstrates the determination of the capacitance C and the inductance L according to the volt-ampere method for alternating current.**

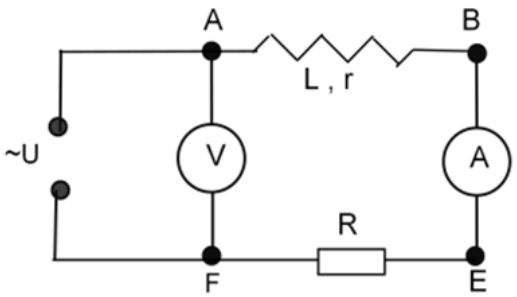


*Figure Lab 1-5: RL circuit schematic*

Suppose an alternating current flowing in the circuit at time $t$ has the form:

$$i = I_o.\sin 2\pi f\,.t \qquad (21)$$

at that time:

$$u = u_R + u_r + u_L \qquad (22)$$

$u_R$ and $u_r$ are in same phase with $i$, $u_L$ is early than $i$ by a phase of $\pi/2$:

$$u = U_{0R}.\sin 2\pi f.t + U_{0r}.\sin 2\pi f.t + U_{0L}.\sin(2\pi f.t - \pi\frac{\pi}{2}) \quad (13)$$

**Question 5: Explain how to determine the absolute error of amperage and voltage measured directly on digital multimeters.**

Set up the experiment with the inductor L and resistor R according to the diagram of Figure Lab 1-5 The AC voltage ~12V is taken from the AC output on the variable power supply. Then use two digital multimeters as an AC voltmeter and an AC ammeter. b. Set the scale for two digital multimeters: The voltmeter V is placed on the ACV 20V AC scale, connected in parallel with the circuits to be measured. Ammeter A placed at 200mA AC scale, two test leads plugged into 2 terminals "COM" and "A", then connected in series to the circuit between R and C by two pins. After the setup is complete, ask the instructor for checking the circuit.

**Question 6: Based on the formula C and L demonstrate the expression for relative error of capacitance C and inductance L form:**

$$C = \frac{I}{2\Pi f U_c}$$

$$lnC = ln(\frac{I}{2\Pi f U_c}) = lnI - ln(2\Pi f U_c) = lnI - ln\Pi - lnf - lnU_c$$

Take the differential and take the absolute of each derivative:

$$\left|\frac{\partial C}{C}\right| = \left|\frac{\partial I}{I}\right| + \left|-\frac{\partial \pi}{\Pi}\right| + \left|-\frac{\partial f}{f}\right| + \left|-\frac{\partial U_c}{U_c}\right|$$

Replace the differential by the uncertainties:

$$\frac{\Delta C}{C} = \frac{\Delta I}{I} + \frac{\Delta \pi}{\pi} + \frac{\Delta f}{f} + \frac{\Delta U_C}{U_C}$$

Same with $\Delta L$

$$L = \frac{\sqrt{Z_{cd}^{\ 2} - r^2}}{2\Pi f}$$

$$ln\, Z = 0.5 ln(Z_{cd}^{\ 2} - r^2) - ln\Pi - lnf - lnU_c$$

$$\left|\frac{\partial Z}{Z}\right| = \left|\frac{Z_{cd}\partial Z_{cd}}{Z_{cd}^{\ 2} - r^2}\right| + \left|\frac{r\partial r}{Z_{cd}^{\ 2} - r^2}\right| + \left|-\frac{\partial \pi}{\Pi}\right| + \left|-\frac{\partial f}{f}\right| + \left|-\frac{\partial U_c}{U_c}\right|$$

$$\frac{\Delta L}{L} = \frac{Z_{cd}\Delta Z_{cd} + r\Delta r}{Z_{cd}^{\ 2} - r^2} + \frac{\Delta \pi}{\pi} + \frac{\Delta f}{f}$$

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